

RESEARCH ARTICLE

The Impact of Working Memory Training on Cognitive Abilities in Older Adults: The Role of Cognitive Reserve

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Abstract: Objectives: The aims of this study were to determine whether working memory (WM) training improves the cognitive functioning of older adults and to determine the role of cognitive reserve in WM training.

Methods: Twenty-one older adults, aged between 65 and 91 years were included in the study. Ten of them were in the experimental group and 11 in the passive control group. The experimental group underwent 15 training sessions of n-back training over a period of five weeks, whereas the control group remained passive. All participants (from the experimental and control group) were tested before the training, one week after the training, and three months after the training with Rey–Osterrieth/Taylor Complex Figure test (ROCF), Digit span, and TMT (part A and part B).

Results and Conclusion: Results of our study suggest that although the experimental group slightly improved their performance on the trained task, the progress was not statistically significant. There was also no statistically significant transfer of training effects onto tasks of visual-spatial and verbal memory, as well as those related to executive functioning. However, the study did identify a statistically significant correlation between cognitive reserve and certain tests performed at the final testing: tasks measuring executive functioning and spatial ability. Results also revealed that the group that showed improvement in the training task was significantly better in the ROCF test in comparison with the group that had not improved their performance on the N-back task. Thus, visual-spatial abilities (visual perception, construction, and memory) were more connected with success in WM training, than other measured cognitive abilities (e.g. verbal and numerical memory).

Keywords: Working memory, cognitive reserve, cognitive training, n-back, executive functions.

1. INTRODUCTION

Cognitive aging is a selective process affecting, for example, working memory and executive functions. The functions that are most affected are those that demand greater mental effort, such as processing speed and the ability to integrate new, complex stimuli [1]. One of the questions currently being addressed in aging research is whether it is possible to preserve older people's level of performance on basic cognitive processing tasks or perhaps even improve it [2]. There is a growing number of studies looking at whether cognitive training can maintain or improve cognitive abilities, but so far the data on the results of training and transfer effects have been mixed and inconclusive [3].

1.1. Working Memory Training with the N-Back Test

It has long been assumed that working memory capacity is limited. However, recent research on working memory training has shown that working memory capacity and

cognitive abilities can be improved by practicing tasks that involve task switching, inhibition and reasoning [4, 5], processing speed [6], and various types of memory span task [2, 7]. Working memory plays an important role in higher-order cognitive processes, which leads to the assumption that working memory training can improve wider cognitive functioning. Recent research findings suggest that the most successful interventions involve divided attention, the ability to manipulate stimuli, and inhibition of irrelevant stimuli. Tasks of this sort are crucial to the successful functioning and improvement of working memory capacity [8].

One of the most frequently used tasks to assess working memory is the n-back task, which requires the participant to assess whether the currently presented stimulus is the same as the one presented n-places previously in a series of stimuli [9]. There are various forms of n-back task, which can be distinguished on the basis of the number and type of stimuli involved. The most commonly used tasks are single and dual n-back tasks. The single n-back task uses stimuli in one modality, whereas the dual n-back task uses two different stimulus modalities simultaneously, visual-spatial and auditory-verbal. The difficulty of the task depends on the value of n.

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The n-back task taps various executive processes: active retention of the last stimulus, updating new stimuli and comparing them with previously presented stimuli [10]. Successful performance requires synchronous activation of all of these processes as well as simultaneous storage and processing of stimulus information [11]. Practicing the n-back task not only helps to improve one's ability to tackle less complicated working memory tasks (*e.g.*, retaining a sequence of numbers), but it also improves performance on executive functioning and verbal fluency tasks [11].

1.2. Efficacy of N-Back Working Memory Training in Elderly People

Investigations of the effect of n-back training on the untrained working memory capacity of older adults have produced inconsistent results. Stepankova *et al.* [5], for example, investigated the effect of adaptive (*i.e.* the difficulty of the n-back task increased with better performance on the task) working memory training with a verbal n-back task on the untrained measures of working memory and visuospatial skills. Healthy older adults (age range: 65-75 years) were trained either for 10 or for 20 sessions over the course of one month. After the training period, both training groups performed significantly better on the trained task as well as on untrained tasks in comparison with the control group.

Similar effects were found by Salminen, Frensch, Strobach, and Schubert [12]. In their study, the dual n-back task was used to train the working memory of older adults (57-73 years) and young adults. The authors compared the effects of training between groups and found that both improved performance of the trained task, but the training effect was smaller in the older adult group. After training the older adults performed at the same level as young adults who had not received any training.

Some studies suggest that working memory training with the n-back task may not improve the performance of other working memory tasks in healthy adults. A study that used the dual n-back task to train a group of middle-aged adults (30- 60 years old) found that after training, the experimental group did not perform any better on other working memory tasks than an active control group [13].

Relative lack of improvement in working memory performance following n-back working memory training may be due to the difficulty of the task. If the training task is too difficult or frustrating, then participants may have trouble reaching their full potential and expanding their abilities [13].

1.3. Transfer of Working Memory Training Effects to Untrained Tasks

Working memory is involved in several cognitive processes and so it is reasonable to assume that an improvement in working memory capacity would be accompanied by an improvement in related cognitive functions. It is assumed that this kind of transfer occurs when the trained task and the transfer task share certain features. However, Buschkuhl *et al.* [6] concluded that common features were not the only prerequisite for transfer. The aim of the training is not to

alter the way participants solve a certain task, but rather to alter how they process information. Successful cognitive training also relies on maintaining a certain level of difficulty whilst taking individual differences into account. This can be achieved by using a task whose difficulty can be adjusted according to the ability of the participant [14].

Borella *et al.* [2] distinguished between “near-transfer” tasks, which share several common features with the trained task, and “far-transfer” tasks, which share fewer features with the trained task and thus are less likely to benefit from the transfer. The same authors examined the impact of verbal working memory training in older adults (65 - 75 years old) and found that the effects transferred to all the tasks they tested, both near- and far-transfer tasks. Moreover, several recent studies have confirmed the transfer of training effects to conceptually close untrained tasks. Basak *et al.* [4] assessed the impact of training - in the form of a complex strategic game that requires constant task-switching and provides individualized feedback - on executive functions and memory processes in older adults. After the training, participants improved their performance in particular on tasks pertaining to executive functions, which indicates near transfer.

Transfer of training effects to untrained tasks does not always occur. Lawlor-Savage and Goghari [13] found that training produced no improvement in working memory outcomes and also no improvement in performance on information processing tasks and fluid intelligence tests.

1.4. Long-Term Effects of Working Memory Training

Like research of short-term transfer effects, studies of the long-term working memory training effects have produced mixed results. Both short-term transfer effects and long-term effects of training are usually limited to tasks that have something in common with the training task and require similar information processing techniques [2]. A study by Li *et al.* [15] observed long-term effects of training (three months after the last training session took place). The study did confirm the long-term effects of training, but only in relation to near-transfer tasks. Similarly, Dahlin *et al.* [8] found long-term effects of training in updating (after three months).

Borella *et al.* [2] examined the effect of training on a verbal memory task on the performance of untrained near- and far-transfer tasks eight months after the training. Older adults who participated in working memory training showed improvements in the performance of the trained task which were maintained eight months later. In contrast, Buschkuhl *et al.* [6] were unable to detect long-term effects of visual working memory training when comparing scores at a session one year after training with those obtained in the initial testing. They did, however, find transfer to untrained conceptually close tasks.

1.5. The Cognitive Reserve Theory

Several older adults may have the same degree of age-related decline in cognitive functioning, but it may not be equally obvious in all of them. One of the explanations given for variability in the manifestation of age-related cognitive

decline depends on the concept of the cognitive reserve [16]. The manifestation of age-related decline is believed to be delayed and reduced in magnitude in people with a higher cognitive reserve, mainly because they adopt alternative strategies for performing certain tasks when the methods they have used hitherto become ineffective [17].

The cognitive reserve hypothesis posits that the brain has the ability to reorganize and thereby compensate for individual deficits in functioning in order to optimize performance [18]. The cognitive reserve can also be defined as the level of cognitive decline that individuals can sustain without it becoming apparent [19]. There are individual differences in the ability to compensate for the onset of cognitive decline. These differences can be ascribed to genetics or differences in life experience, such as level of education, professional demands, and leisure activities [19]. In that vein, Wajman, Mansur, and Yassuda [20] emphasized that older people who regularly participate in different social, cultural, and intellectual activities tend to perform better on cognitive tests, manifest fewer cognitive complaints, and are less likely to develop neurodegenerative disorders. Cognitive reserve is also the ability of the adult brain to optimize cognitive functioning by (1) recruiting new brain areas and (2) adopting alternative cognitive strategies to compensate for an impairment [21].

There is a hypothesis that cognitive reserve may enhance response to cognitive training since individuals with a greater cognitive reserve achieve better scores in cognitive training than those with a smaller cognitive reserve [22].

Calero and Navarro [22] also sought to demonstrate a correlation between cognitive reserve and memory training scores. Their findings confirmed that memory training improves cognitive performance and also confirmed the existence of a positive association between cognitive reserve and performance on trained tasks. These findings attest to the importance of cognitive reserve as a potential explanation for the immediate and long-term benefits of memory training for older adults [22].

1.6. Aim of the Study

This study examined the potential impact of working memory training on cognitive functions in older adults. We were particularly interested in the effects of working memory training on performance of the trained n-back task and potential transfer of training effects to other areas of cognitive functioning (visuospatial memory; verbal memory; numerical memory; executive functions). Another research question was whether the effects of working memory training depend on the magnitude of an individual's cognitive reserve.

2. METHODS

2.1. Participants

The participants were elderly adults (≥ 65 years) living in a retirement home. Twenty-nine potential participants were screened using the Mini-Mental State Examination (MMSE) and 24 met the inclusion criteria. Individuals who scored fewer than 21 points (the criterion for moderate to severe

cognitive decline) on the MMSE were not included in the study. Other exclusion criteria were uncorrected impaired sight or hearing, upper limb mobility problems, the presence of psychiatric or neurological disorders, head injury, and use of medication that could affect cognitive performance. Another three participants were later excluded from the study because they did not finish the cognitive training.

The final sample comprised 21 elderly individuals (15 women) between 65 and 91 years of age ($M = 81.71$, $SD = 7.14$). The control group comprised of 11 participants, aged between 65 and 91 years of age ($M = 82.82$, $SD = 7.88$) and the training group comprised of 10 participants, aged between 70 and 89 years of age ($M = 80.50$ years; $SD = 6.40$). The mean ages of the both groups did not differ, $t(19) = 0.74$; $p = .47$. There was no statistically significant difference in cognitive reserve index for experimental group and control group, $t(19) = -0.28$; $p = .79$ (see Table 2 for descriptive statistics).

2.2. Instruments

Cognitive Reserve Index (CRI). The CRI [21] is an instrument used to assess an individual's cognitive reserve based on information about his or her education, career, and spare-time activities. The first part of the questionnaire deals with education, capturing information about years of formal education and occupational training. The second part contains questions about the types of work in which the individual was engaged in during his or her working life, as well as the length of time spent in employment. Jobs are categorized into five types, based on the cognitive demand and the level of responsibility they entail. The third part of the questionnaire examines spare-time activities and includes questions about cognitively stimulating activities that the individual conducts or conducted during his or her free time.

Training task: The single n-back task (a modified form of the task used by Jaeggi *et al.*, [14]) was administered using the computerized "Brain Workshop" app. The n-back task requires participants to decide whether the current stimulus is the same as the stimulus presented n places previously. The single n-back task presents stimuli of just one type, in our case visuospatial stimuli. At the start of the task participants are asked to decide whether the current stimulus is the same as the previous stimulus ($n = 1$) and n is then increased or decreased depending on the accuracy of the participant's performance over the previous three trials (increased if accuracy $\geq 80\%$; decreased if accuracy $< 50\%$).

Transfer tasks: The Rey-Osterrieth complex figure test (Osterrieth, 1944; Rey, 1941 [23]) is a mnemonic test of visual perception that consists of several components making up a complex abstract figure. Participants are asked to copy the image onto a blank sheet of paper to the best of their ability. After a few minutes, they are asked to redraw the image from memory. Approximately 30 minutes later (after solving the remaining tasks) they are once again asked to redraw the figure. The test assesses visual and spatial capabilities, construction capabilities, planning, memory, and attention [23].

Transfer tasks: Taylor's complex figure test (Taylor, 1969 [24]) is a modified form of the Rey-Osterrieth test

and is often used in tandem with it in test-retest studies to limit the memory effect. The complex figure used in the Taylor test comprises the same number of components as that in the Rey-Osterrieth test, but they are presented differently [24].

Transfer tasks: The digit span test (Wechsler, 1997 [25]) is one of the most commonly used neuropsychological tests of verbal and numeric memory [26]. Participants are asked to recall a series of digits in the order in which they were presented (forward digit span) or in the reverse order (backward digit span). The length of the series increases from 3 to 9 in the forward digit span test and from 2 to 8 in the backward digit span test. The test ends when the participant makes two mistakes in a row or when they successfully repeat the maximum number of digits.

Transfer tasks: The trail-making test (TMT; Partington & Leiter, 1949 [27]) is one of the most commonly used tests for assessing executive functions in adults. The TMT provides measures of psychomotor speed, complex attention, the speed of visual perception and mental flexibility (Corrigan & Hinkeldey, 1987 [27]). It consists of two parts. The first part (TMT A) is a task in which participants must connect circles marked from 1 to 25 in the correct order and the second part requires participants to alternate between linking circles marked 1 to 12 in ascending order and circles marked with the letters A to L in alphabetical order. The second part of the test thus requires participants to switch repeatedly between tasks.

2.3. Procedure

Screening sessions (questionnaire and the MMSE) were conducted individually during the first week of the study. The screening was followed by the baseline assessment (T1), which consisted of the CRI, which was administered in an interview, the Rey-Osterrieth complex figure test, the digit span test, and the TMT. Screening sessions, training and all assessments (T1, T2, and T3) were conducted individually in participant's rooms. All procedures were carried out with the approval of the Republic of Slovenia national medical ethics committee and written informed consent was obtained from all participants.

After the baseline assessment, participants were divided into experimental and passive control groups based on their willingness to participate in cognitive training. The assignment into two groups was not random. Because of our sample specifics, we included only participants that were willing to participate in the training condition. The experimental group received memory training using a single n-back task. Training took place three times weekly for five weeks (15 sessions altogether). Each session lasted approximately 20 minutes. Training sessions were conducted individually. We used the adaptive form of the n-back task, where the difficulty of the task is adjusted according to the participant's in-task performance. The adaptive form makes training more efficient as it enables individuals to make the best use of their potential [28].

One week after the final training session, the Taylor complex figure test, digit span test, and the TMT were once again administered to both groups (T2). Finally, three

months after the post-training assessment we carried out a repeated assessment (T3) in order to assess potential long-term effects of memory training using n-back tasks. The same tests were employed as in the T2 assessment.

2.4. Statistical Analysis

Statistical analyses were performed using IBM SPSS Statistics version 24. Group differences in progress on the n-back task were assessed with Friedman's ANOVA and paired samples *t*-tests. Group differences in performance on the untrained tasks in all three assessments (T1, T2, T3) were examined using paired samples *t*-tests and repeated measures ANOVA. The effect of cognitive reserve on the performance of untrained tasks was evaluated with ANCOVA.

3. RESULTS

3.1. Progress on the Trained Task

Friedman test revealed no significant differences in the n-back value between sessions, ($\chi^2(14) = 22.50$; $p = .07$), meaning that the value of *n* (the maximum number of places back for which the participant could correctly judge whether a previously presented stimulus was the same as the current stimulus) was similar in all sessions. However, there was a trend towards improvement (see Table 1 for details). The highest achieved level (by 4 participants) was 3-back; 4 participants achieved the level of 2-back, and 2 participants did not improve from the initial 1-back level. The age difference between those that showed improvement on the trained task ($M = 77.25$ years) and those that did not show any improvement ($M = 82.76$ years), $t(8) = 1.38$, $p = .21$ was not statistically significant.

3.2. Transfer on Untrained Tasks and Long-term Effects

The results showed no group differences in the performance of the tasks at the baseline assessment ($p > .05$ for all tests). Descriptive statistics for all measurement points are shown in Table 2.

There was no statistically significant interaction effect between the scores on the Rey/Taylor test - copying and group (experimental and control group) in T1 and T2. The average score on Rey/Taylor -copying test was slightly higher in the post-training assessment (T2) than in the baseline assessment (T1) for both groups. The results were similar for Rey/Taylor - immediate recall and Rey/Taylor - delayed recall, although here the differences between T1 and T2 were to some extent larger for the experimental group. (see Fig. 1 and Table 1 in supplement).

In addition, we divided the experimental group into two subgroups; the one which improved on the trained task ($N = 4$) and the one with no improvement ($N = 6$). The results showed that those who improved on the trained task performed significantly better on the Rey-Osterrieth/ Taylor complex figure test. The difference between groups was marginally significant for copying ($p = .05$) and strongly statistically significant for immediate recall ($p = .005$) and delayed recall ($p = .01$). The results on other tests did not differentiate between groups.

Table 1. Requery of distribution of n-back values and basic descriptive statistics for results in n-back tests.

Training	n-back				
	1-back	2-back	3-back		
	<i>f</i>	<i>f</i>	<i>f</i>	<i>M</i>	<i>SD</i>
1	4	5	1	1.70	0.67
2	3	7	-	1.70	0.48
3	3	7	-	1.70	0.48
4	2	8	-	1.80	0.42
5	3	7	-	1.70	0.48
6	3	7	-	1.70	0.48
7	2	7	1	1.90	0.57
8	3	5	2	1.90	0.74
9	2	7	1	1.90	0.57
10	3	6	1	1.80	0.63
11	2	7	1	1.90	0.57
12	2	7	1	1.90	0.57
13	2	7	1	1.90	0.57
14	2	7	1	1.90	0.57
15	2	4	4	2.20	0.79

Notes. *M* = arithmetic mean; *SD* = standard deviation.

Table 2. Reapeted measures ANOVA for second (T2) and repeated assessment (T3).

-	CRI	Rey / Taylor (Copying)	Rey / Taylor (Im- mediate Recall)	Rey / Taylor (Delayed Recall)	Span (For- ward)	Span (Back- ward)	TMT A	TMT B
T 1								
Control group	95.73 (11.78)	33.82 (10.72)	14.31 (3.65)	11.81 (4.78)	5.27 (1.10)	3.18 (1.33)	100.00 (45.07)	166.00 (67.88)
<i>n</i>	11	11	11	11	11	11	11	6
Experimental group	97.30 (14.38)	30.50 (5.95)	13.50 (7.48)	11.81 (8.41)	5.20 (0.92)	3.10 (1.29)	99.90 (73.34)	176.67 (79.51)
<i>n</i>	10	10	10	10	10	10	10	5
T 2								
Control group	-	39.64 (4.64)	15.94 (5.31)	12.25 (4.79)	5.82 (0.87)	3.45 (1.04)	83.94 (48.88)	199.00 (97.58)
<i>n</i>	-	11	8	8	11	11	10	9
Experimental group	-	38.10 (5.60)	18.00 (8.87)	17.19 (10.21)	5.20 (1.03)	3.30 (0.68)	62.35 (36.91)	142.00 (56.67)
<i>n</i>	-	10	9	8	10	10	10	6
T 3								
Control group	-	36.86 (5.76)	15.21 (4.93)	14.79 (5.12)	6.00 (1.41)	3.86 (1.07)	96.67 (39.08)	230.50 (38.89)
<i>n</i>	-	7	7	7	7	7	6	4
Experimental group	-	32.10 (5.70)	17.17 (9.00)	14.75 (6.59)	5.00 (1.10)	3.67 (1.37)	76.00 (49.64)	135.00 (5.66)
<i>n</i>	-	6	6	6	6	6	5	4

Means, standard deviations *M*(*SD*) and *n* for CRI, Rey-Osterrieth / Taylor complex figure test, Digit span test forward and backward, Trail making test A and B for all three assessments (T1, T2 and T3).

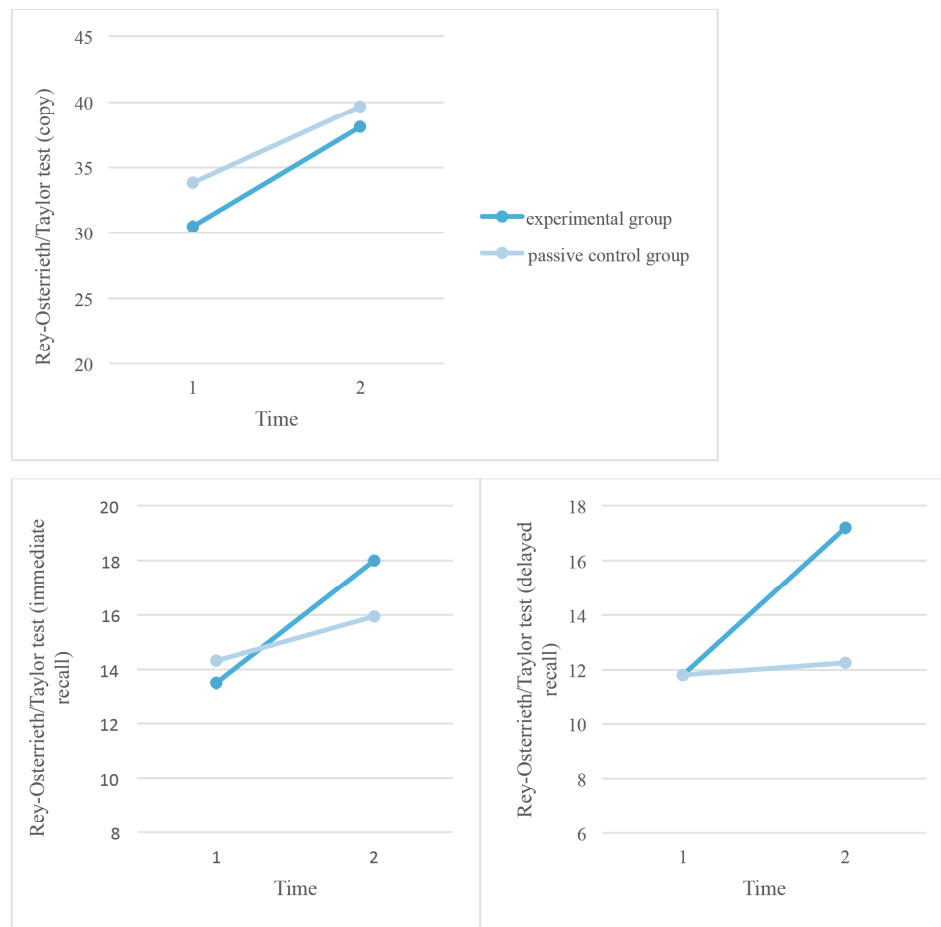


Fig. (1). Mean values on the Rey-Osterrieth/Taylor complex figure test (copying, immediate recall and delayed recall) at the initial (T1) and final (T2) testing for the control and experimental group.

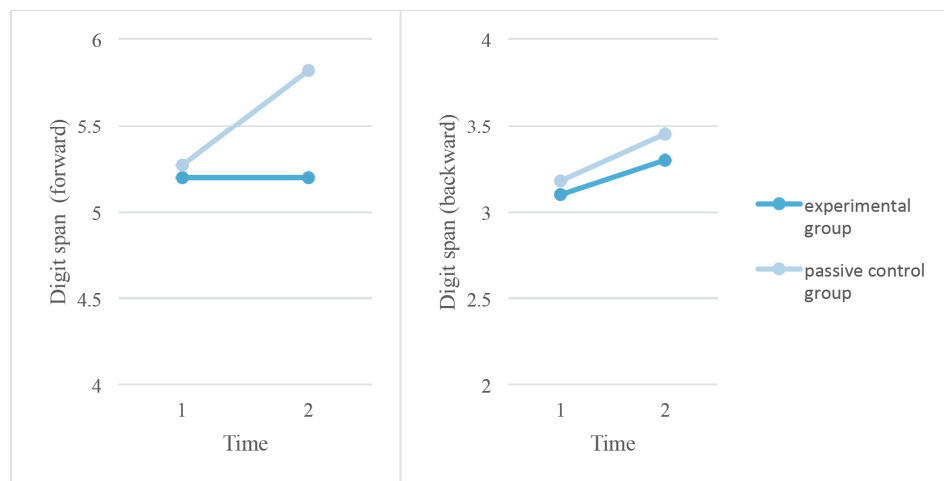


Fig. (2). The average values on the symbol span test forward and backward in initial (T1) and final (T2) testing for the control and experimental groups.

The interactions between the score on the digit span test (forward and backward) and the group were not statistically significant (see Table 1 in supplement). However, the control group had slightly higher average results at both assessments (T1 and T2) (Fig. 2).

There was also no statistically significant interaction effect between the score on TMT A/ TMT B and group in a

comparison between baseline (T1) and post-training assessment (T2) (see Table 1 in supplement). On TMT A, both groups achieved a better average time in the final assessment (T2), as shown in Fig. (3). However, on TMT B, the experimental group achieved better average time in the post-training assessment (T2) in comparison to baseline assess-

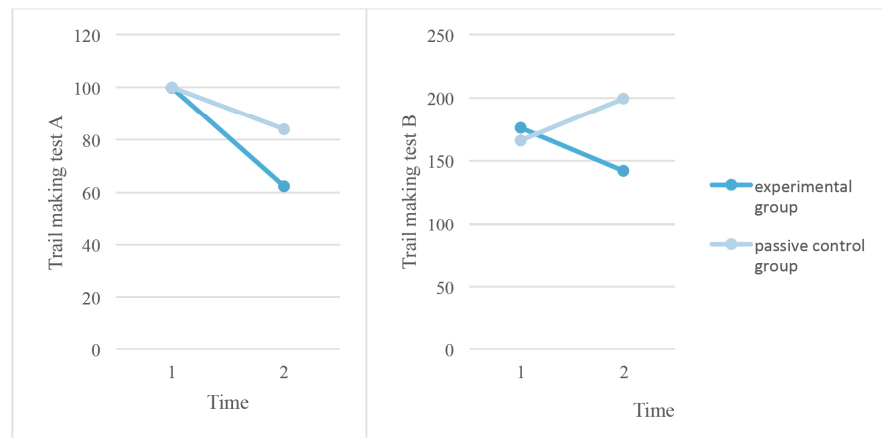


Fig. (3). The average values on the trail making test (TMT A and TMT B) in initial (T1) and final (T2) testing for the control and experimental groups.

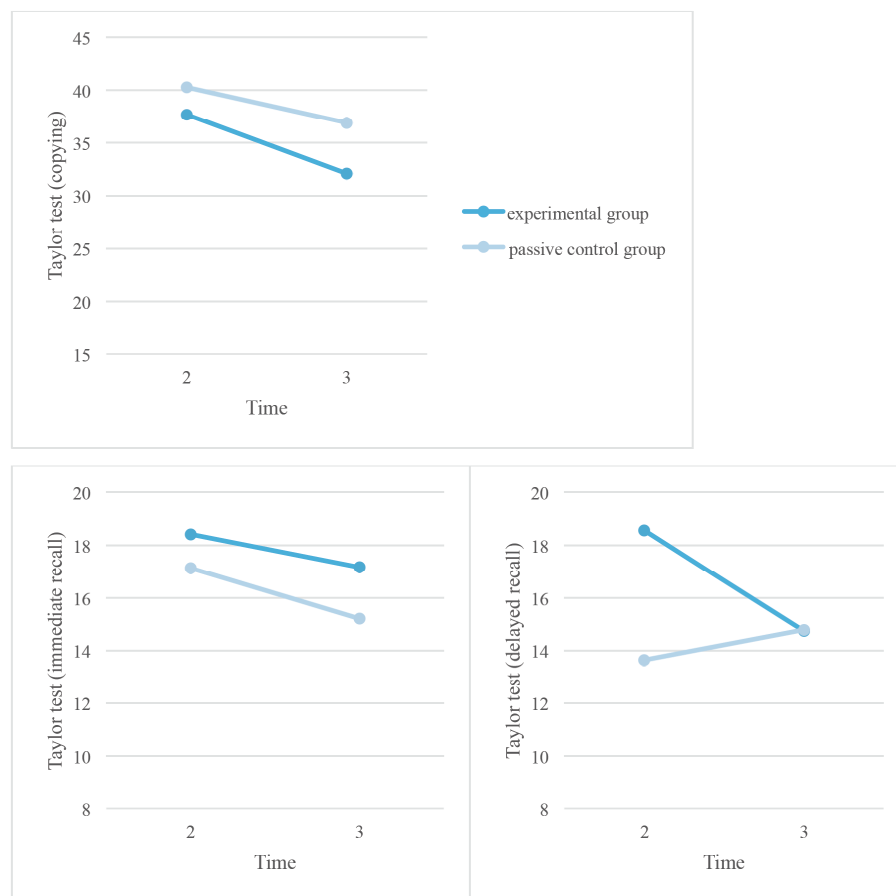


Fig. (4). Mean values on the Rey-Osterrieth/Taylor complex figure test (copying, immediate recall and delayed recall) at the final (T2) and repeated (T3) testing for the control and experimental group.

ment (T1) (Fig. 3), which was not the case with the control group.

In comparison of scores on Taylor test (copying) for T2 and T3, separate repeated measures ANOVAs for each group revealed an effect of time on performance, for the experimental group ($p = .002$) as well as for the control group ($p = .03$). Both groups achieved a better result in post-training assessment (T2) than on repeated testing (T3) (Fig. 4). Similar results were shown for Taylor test (immediate recall), at the post-training (T2) and repeated (T3) assessment. In de-

layed recall, however, the experimental group performed better on Taylor test at the post-training assessment (T2) compared to repeated assessment T3, whereas the control group showed an opposite pattern (see Fig. (4) and Table 2 in supplement).

There was no statistically significant interaction effect between the score on Digit span test (forward) and group at the post-training (T2) assessment and repeated (T3) assessment (see Table 2 in supplement). Surprisingly, both groups

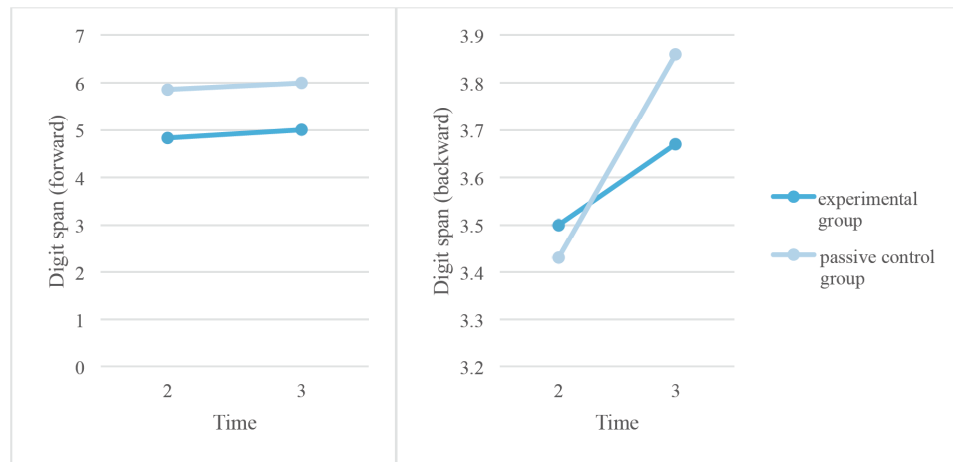


Fig. (5). The average values on the symbol span test forward and backward in final (T2) and repeated (T3) testing for the control and experimental groups.

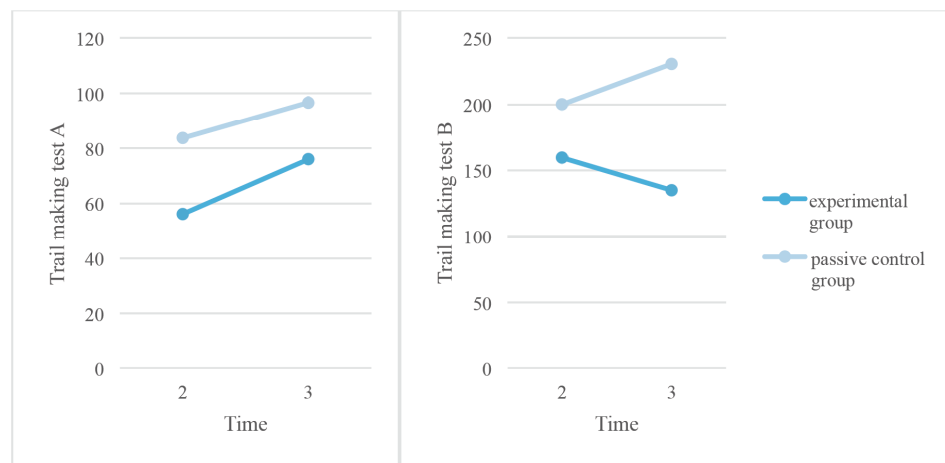


Fig. (6). The average values on the trail making test - TMT A and TMT B in final (T2) and repeated (T3) testing for the control and experimental groups.

achieved better results in the Digit span test (backward) at the repeated (T3) assessment in comparison with T2 (Fig. 5).

Also, the interaction between the average score on TMT A / TMT B and group was not statistically significant (Table 2 in supplement). Both groups needed slightly more time to complete the TMT A task in the repeated (T3) assessment in comparison to post-training (T2) assessment; in TMT B, the experimental group was faster in the repeated assessment (Fig. 6).

3.3. The Effect of Cognitive Reserve on Task Performance

We were interested in whether cognitive reserve differs between those who showed an improvement in the trained task and those with no improvement. Results showed no difference in cognitive reserve between the group with improvement ($M = 97.25$) and without an improvement on the trained task ($M = 97.33$), $t(8) = 0.01$, $p = .99$. However, including cognitive reserve as a covariate did affect the results on Taylor complex figure test ($F(1,18) = 4.20$; $p = .05$; $r = .43$) and TMT B task ($F(1,8) = 7.30$; $p = .03$; $r = .69$) at post-training measurement T2.

4. DISCUSSION

Our aims were to determine whether working memory training improves performance on trained and untrained cognitive tasks and to assess the possible effect of cognitive reserve in response to training. Results showed that participants who trained working memory with the n-back task did not significantly improve their performance on this task, although there was a trend towards an improvement in n over the course of individual training sessions. It was also shown that cognitive reserve influenced post-training (T2) performance of the executive functioning task (TMT B) and the Taylor complex figure task.

Although the results failed to show significant progress on the trained task for the experimental group, some participants improved their performance and achieved the level of 3-back while others did not show any improvement during the training. These findings are in accordance with earlier studies showing that although elderly people do have the capacity to improve their performance on trained tasks, that capacity is limited and is somewhat smaller than in younger adults [8]. Similarly, we found that the mean age of those that showed improvement in the trained task was slightly lower in comparison with the group that did not show any

improvement, which also coincides with the previous research findings. Interestingly, the results also showed that there was a statistically significant difference in Rey-Osterrieth / Taylor complex figure test between these two groups: the group that showed improvement on the training task was significantly better in the ROCF test in comparison with the group that has not improved their performance on the N-back task. Visual-spatial abilities (visual perception, construction, planning, and memory), which are also very age-sensitive, were obviously somehow more connected with success in WM training, than other measured cognitive abilities (e.g. verbal and numerical memory).

Additionally, cognitive plasticity decreases with age [29], which is most likely the reason for the limited effectiveness of training in this sample of elderly adults. Younger adults have a greater potential to improve their working memory through training, and on this basis it has been suggested that working memory training in the early years of old age (before the average age of onset for age-related cognitive decline) might be crucial to delaying the onset of other forms of cognitive decline [13]. The lack of training effect in our sample may also be due to the type of training or the difficulty of the trained task. It may be that the n-back task was too difficult for our elderly participants and thus prevented them from realizing a maximal benefit from training. Although the n-back test is flexible: the difficulty level can be adjusted according to individual performance, increasing the difficulty of the task immediately after a short period of good performance can reduce the motivating impact of success. Lawlor-Savage and Goghari [13] found that motivation levels were directly negatively affected when good performance was rewarded with a more difficult task.

Our hypothesis that progress on the trained task would be linked to cognitive reserve, hinges on the assumption that the brain is capable of reorganizing to compensate for age-related cognitive deficits [18]. We predicted that participants with a greater capacity for compensation would benefit more from training. According to Baltes [30], there exists a capacity, termed cognitive reserve, which represents the maximum potential of an individual's cognitive system and can be activated under optimal conditions, which cognitive training can provide.

As assumed, we found the effect of cognitive reserve on post-training (T2) performance of the Taylor complex figure test and TMT B. We can infer from these results that cognitive reserve may influence the capacity for transfer of working memory training to untrained tasks, specifically those that tap executive functioning and spatial abilities. Also, as we stated before, visual-spatial abilities, which are also very age-sensitive, are according to our results more connected with success in WM training, than other measured cognitive abilities.

The results of studies that examined transfer onto untrained tasks are also inconsistent, indicating that transfer of training effects to untrained tasks does not always occur and is probably dependent on certain conditions. In our study, performance on the untrained task did not significantly improve in the experimental in comparison with the control group. This is probably because the experimental group did not make sufficient progress on the trained task since this is a

prerequisite for the transfer of training to untrained tasks. Also, the limited effectiveness of n-back training in this study could be possibly ascribed to elderly people's lack of familiarity with computerized tasks. In our sample, 17 from 21 participants reported, that they use computers never or very rare. Future studies should investigate whether older adults with more computer-experience adapt faster to computerized training in comparison with older adults with no experiences. The important limitation of our study was the lack of random assignment into the experimental and passive control group. It is also possible that the fact, that post-training assessment (T2) was conducted one week after the final training session may influence the results, therefore the future studies may consider conducting the testing immediately after the end of the training period. In the end, it is very possible that the n-back training would have a significant positive effect on cognitive functions also in this specific group of older adults - under different conditions, e.g. more sessions, longer sessions, more frequent training, etc.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

Not applicable.

HUMAN AND ANIMAL RIGHTS

No Animals/Humans were used for studies that are base of this research.

CONSENT FOR PUBLICATION

Not applicable.

AVAILABILITY OF DATA AND MATERIALS

Not applicable.

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CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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SUPPLEMENTARY MATERIAL

Supplementary material is available on the publisher's web site along with the published article.

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